Never Accept the Null Hypothesis

Consider a game of Roulette and wagering a bet on the ball landing in an even slot.

Suppose you play 20 times and win 8 times (or 40% of the games).

With this bet, the green slots are reserved for the house, so the actual probability of winning is 18/38, or 47.4% of the time.

With your sample, did you lose less than half the time?

Set up a simulation for the following hypothesis:

$$H_0: \pi = 0.5$$
$$H_A: \pi < 0.5$$

Under “Count Samples” in the applet, enter as follows:

As extreme as ≤ 0.4 (if “Proportion of heads” is your stat.)

Your resulting p-value is greater than 0.05, so we would fail to reject our null hypothesis. However, the true long-run proportion of winning a bet with even numbers is 47.4%, not 50% (all casino games are tilted slightly in the house’s favor).

In this case, we know the parameter because of probability, but this is not a luxury in real world studies.

Since we typically do not know the parameter in populations, <u>we cannot accept the null hypothesis</u>.

Therefore, we either reject or fail to reject the null hypothesis.

In this case, never say “We accept the null that placing a bet on even numbers will win 50% of the time”

Conclusion: “We do not have evidence to conclude that the long run proportion of times winning roulette by placing a bet on even number is less than 0.5.”